

Supervised Machine Learning Final Exam

120 Practice Questions with Correct Answers | Complete Question Bank

Module 1: Introduction to Statistical Learning & ML Fundamentals

ISLR 2.1–2.3, Video Lectures SL1–SL4, Lab 1

Question 1: What is the primary distinguishing characteristic between supervised learning and unsupervised learning?

- A. Supervised learning requires continuous target variables, whereas unsupervised learning requires categorical targets.
- B. Supervised learning relies on labeled training data containing both inputs and target outputs, whereas unsupervised learning uses data without target outputs.
- C. Unsupervised learning uses parametric models, whereas supervised learning always uses non-parametric models.
- D. Supervised learning is used only for inference, while unsupervised learning is used only for prediction.

Correct Answer: B. Supervised learning relies on labeled training data containing both inputs and target outputs, whereas unsupervised learning uses data without target outputs.

Question 2: In the statistical learning framework $Y = f(X) + \epsilon$, what does the error term ϵ represent?

- A. The reducible error that can be eliminated by choosing a more flexible model.
- B. The systematic linear relationship between the predictors and the response.
- C. The irreducible error containing unmeasured variables and random noise, having mean zero and variance σ^2 .
- D. The parameter estimation error resulting from a small sample size.

Correct Answer: C. The irreducible error containing unmeasured variables and random noise, having mean zero and variance σ^2 .

Question 3: A financial analyst wants to determine which macroeconomic factors (interest rates, GDP growth, inflation) significantly influence stock market returns and by how much. What type of task is this?

- A. Unsupervised clustering
- B. Prediction focused on a black-box model
- C. Inference on a supervised learning model
- D. Non-parametric density estimation

Correct Answer: C. Inference on a supervised learning model

Question 4: Which of the following is a key advantage of parametric statistical learning methods over non-parametric methods?

- A. Parametric methods can perfectly fit any arbitrary, highly non-linear functional form without bias.
- B. Parametric methods reduce the task of estimating $f(X)$ to estimating a fixed, finite set of parameters, making them simpler and requiring less data.
- C. Parametric methods completely eliminate the risk of overfitting.
- D. Parametric methods do not require making any functional assumptions about $f(X)$.

Correct Answer: B. Parametric methods reduce the task of estimating $f(X)$ to estimating a fixed, finite set of parameters, making them simpler and requiring less data.

Question 5: How does increasing a model's flexibility generally affect its interpretability?

- A. As flexibility increases, interpretability increases.
- B. As flexibility increases, interpretability decreases.
- C. Flexibility and interpretability are completely independent.
- D. Flexibility decreases interpretability only for linear models.

Correct Answer: B. As flexibility increases, interpretability decreases.

Question 6: In a regression setting, how is the quality of fit typically measured on a dataset?

- A. Using the misclassification rate.
- B. Using the Mean Squared Error (MSE), which averages the squared differences between observed and predicted values.
- C. Using the total variance of the predictor matrix X .
- D. Using the area under the ROC curve (AUC).

Correct Answer: B. Using the Mean Squared Error (MSE), which averages the squared differences between observed and predicted values.

Question 7: Why is a model with an extremely low training MSE not guaranteed to perform well on new, unseen test data?

- A. Training MSE is always larger than test MSE.
- B. The model may have overfit the training data by fitting random noise rather than the true underlying signal $f(X)$.
- C. Training MSE measures bias, whereas test MSE measures irreducible error only.
- D. High flexibility automatically increases training MSE.

Correct Answer: B. The model may have overfit the training data by fitting random noise rather than the true underlying signal $f(X)$.

Question 8: In the mathematical decomposition of expected test MSE, what are the three fundamental components?

- A. Training error, test error, and cross-validation error.
- B. Squared bias of $\hat{f}(x_0)$, variance of $\hat{f}(x_0)$, and variance of the irreducible error ϵ .
- C. Intercept variance, slope variance, and R^2 score.
- D. Precision, recall, and F1-score.

Correct Answer: B. Squared bias of $\hat{f}(x_0)$, variance of $\hat{f}(x_0)$, and variance of the irreducible error ϵ .

Question 9: How are model bias and model variance defined in machine learning?

- A. Bias is the error from sensitivity to training sample fluctuations; variance is error from simplifying assumptions.
- B. Bias is the systematic error introduced by approximating a complex real-world problem with a simpler model; variance is the amount by which $\hat{f}$ would change if estimated using a different training set.
- C. Bias measures test error; variance measures training error.
- D. Both bias and variance decrease monotonically as model flexibility increases.

Correct Answer: B. Bias is the systematic error introduced by approximating a complex real-world problem with a simpler model; variance is the amount by which $\hat{f}$ would change if estimated using a different training set.

Question 10: Which scenario best describes a model suffering from underfitting (high bias)?

- A. A complex 15th-degree polynomial that yields 0 training error but massive test error.
- B. A simple linear regression line fitted to data generated from a highly curved, non-linear function, yielding poor performance on both training and test data.
- C. A K-nearest neighbor model with $K=1$ fitted to a large dataset.
- D. A decision tree grown until every terminal node contains exactly one observation.

Correct Answer: B. A simple linear regression line fitted to data generated from a highly curved, non-linear function, yielding poor performance on both training and test data.

Question 11: In the classification setting, what is the Bayes decision boundary?

- A. A linear regression plane fitted to binary 0/1 outcomes.
- B. The optimal boundary that assigns an observation to the class for which its conditional probability $P(Y=k|X=x_0)$ is maximized.
- C. The boundary formed by setting the training error rate to zero.
- D. The cutpoint determined by minimizing residual sum of squares in regression trees.

Correct Answer: B. The optimal boundary that assigns an observation to the class for which its conditional probability $P(Y=k|X=x_0)$ is maximized.

Question 12: Why is the Bayes error rate generally greater than zero?

- A. Because classification models always underfit the training data.
- B. Because the true population classes may overlap, creating irreducible error in $P(Y|X)$.
- C. Because K-nearest neighbors cannot approximate non-linear boundaries.
- D. Because training sets are always smaller than test sets.

Correct Answer: B. Because the true population classes may overlap, creating irreducible error in $P(Y|X)$.

Question 13: What is the primary limitation of irreducible noise $\text{Var}(\epsilon)$ in predictive modeling?

- A. It can be completely removed using regularized polynomial regression.
- B. It sets an absolute theoretical lower bound on expected test error that no model can overcome.
- C. It decreases as the sample size n approaches infinity.
- D. It only affects classification models and not regression models.

Correct Answer: B. It sets an absolute theoretical lower bound on expected test error that no model can overcome.

Question 14: In Python's `scikit-learn` library, what is the standard workflow sequence for preparing and fitting a supervised learning estimator?

- A. ``predict()`` ``to`` ``fit()`` ``to`` ``score()``
- B. ``fit()`` on training data ``to`` ``predict()`` on test data ``to`` evaluate performance metrics
- C. ``transform()`` ``to`` ``predict()`` ``to`` ``fit()``
- D. ``evaluate()`` ``to`` ``fit_transform()`` ``to`` ``train()``

Correct Answer: B. ``fit()`` on training data ``to`` ``predict()`` on test data ``to`` evaluate performance metrics

Question 15: Suppose you have a dataset with $n = 30$ observations and $p = 100$ features. Which statistical learning approach is most prone to severe overfitting if applied directly without regularization?

- A. Simple linear regression with 1 predictor
- B. Unregularized Ordinary Least Squares multiple linear regression
- C. Ridge regression with a large penalty λ
- D. K-nearest neighbors with $K=25$

Correct Answer: B. Unregularized Ordinary Least Squares multiple linear regression

Module 2: Simple & Multiple Linear Regression

ISLR 3.1–3.2, Video Lectures SL5–SL8, Labs 1 & 2

Question 16: What criterion does Ordinary Least Squares (OLS) use to choose the coefficient estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ in simple linear regression?

- A. Maximizing the Total Sum of Squares (TSS).
- B. Minimizing the Residual Sum of Squares (RSS).
- C. Minimizing the sum of absolute errors.
- D. Maximizing the standard error of the slope.

Correct Answer: B. Minimizing the Residual Sum of Squares (RSS).

Question 17: Which mathematical property is guaranteed to hold for the simple linear regression line fitted by Ordinary Least Squares?

- A. It passes through the origin (0, 0).
- B. It passes through the point of sample means $(\bar{x}, \bar{y})$.
- C. Its slope is always equal to 1.0.
- D. It yields an R^2 of exactly 1.0.

Correct Answer: B. It passes through the point of sample means $(\bar{x}, \bar{y})$.

Question 18: In a simple linear regression model predicting sales (Y, in thousands) from TV advertising (X, in thousands of dollars), $\hat{\beta}_1 = 0.0475$. How is this coefficient interpreted?

- A. An additional 1,000 spent on TV advertising is associated with an average increase in sales of 47.5 units.
- B. An additional 1 spent on TV advertising guarantees an increase in sales of 47.5 units.
- C. TV advertising accounts for 4.75% of total sales.
- D. Sales increase by 47.5% whenever TV advertising doubles.

Correct Answer: A. An additional 1,000 spent on TV advertising is associated with an average increase in sales of 47.5 units.

Question 19: In testing $H_0: \beta_1 = 0$ versus $H_a: \beta_1 \neq 0$, what does a very small p-value (e.g., $p < 0.0001$) indicate?

- A. There is strong evidence that no relationship exists between X and Y.
- B. There is strong evidence to reject H_0 , concluding that a statistically significant linear relationship exists between X and Y.
- C. The model explains 100% of the variance in Y.
- D. The predictor X is collinear with the intercept.

Correct Answer: B. There is strong evidence to reject H_0 , concluding that a statistically significant linear relationship exists between X and Y.

Question 20: What does the R^2 statistic measure in linear regression?

- A. The absolute standard deviation of the error terms ϵ .
- B. The proportion of total variance in the response Y that is explained by the linear model.
- C. The probability that the true slope β_1 equals zero.
- D. The total number of parameters in the model.

Correct Answer: B. The proportion of total variance in the response Y that is explained by the linear model.

Question 21: In simple linear regression, what is the exact mathematical relationship between the sample correlation coefficient $r = \text{Corr}(X, Y)$ and the R^2 statistic?

- A. $R^2 = r$
- B. $R^2 = r^2$
- C. $R^2 = 1 - r$
- D. $R^2 = \sqrt{r}$

Correct Answer: B. $R^2 = r^2$

Question 22: What is the primary conceptual advantage of multiple linear regression over running separate simple linear regressions for each predictor?

- A. Multiple regression avoids using matrix algebra.
- B. Multiple regression isolates the unique effect of each predictor on the response while holding all other predictors constant.
- C. Multiple regression guarantees that all p-values will be significant.
- D. Multiple regression requires fewer observations than predictors ($n < p$).

Correct Answer: B. Multiple regression isolates the unique effect of each predictor on the response while holding all other predictors constant.

Question 23: In a multiple linear regression model $\text{Sales} = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \text{Newspaper}$, why might the coefficient for Newspaper be statistically significant in a simple regression but insignificant in the multiple regression?

- A. Newspaper budget is negatively correlated with Sales.
- B. Newspaper spending is correlated with Radio spending; Radio drives sales, so Newspaper was acting as a surrogate in simple regression (confounding).
- C. Multiple regression uses t-tests instead of F-tests.
- D. The sample size n decreased in multiple regression.

Correct Answer: B. Newspaper spending is correlated with Radio spending; Radio drives sales, so Newspaper was acting as a surrogate in simple regression (confounding).

Question 24: What is the null hypothesis evaluated by the overall F-statistic in multiple linear regression with p predictors?

- A. $H_0: \beta_0 = 0$
- B. $H_0: \beta_1 = \beta_2 = \dots = \beta_p = 0$
- C. $H_0: R^2 = 1.0$
- D. $H_0: \text{RSS} = 0$

Correct Answer: B. $H_0: \beta_1 = \beta_2 = \dots = \beta_p = 0$

Question 25: Under what condition is the overall F-test particularly essential over looking at individual t-tests?

- A. When $p = 1$.
- B. When p is large, because looking at individual t-tests at $\alpha = 0.05$ creates a high false positive risk (5% chance per test of falsely rejecting a true null hypothesis).
- C. When R^2 is exactly 0.5.
- D. When all predictors are qualitative.

Correct Answer: B. When p is large, because looking at individual t-tests at $\alpha = 0.05$ creates a high false positive risk (5% chance per test of falsely rejecting a true null hypothesis).

Question 26: What happens to the training R^2 when an additional predictor variable is added to a multiple linear regression model?

- A. Training R^2 always decreases.
- B. Training R^2 will always increase or remain the same, even if the added variable is completely unrelated to the response.
- C. Training R^2 becomes equal to Adjusted R^2 .
- D. Training R^2 drops to zero if the feature is collinear.

Correct Answer: B. Training R^2 will always increase or remain the same, even if the added variable is completely unrelated to the response.

Question 27: What is the primary advantage of the closed-form Normal Equations in solving Ordinary Least Squares multiple linear regression?

- A. They require no matrix inversions.
- B. They provide a direct, non-iterative mathematical solution: $\hat{\beta} = (X^T X)^{-1} X^T Y$.
- C. They work even when predictors are perfectly collinear.
- D. They automatically scale all input features between 0 and 1.

Correct Answer: B. They provide a direct, non-iterative mathematical solution: $\hat{\beta} = (X^T X)^{-1} X^T Y$.

Question 28: In scikit-learn or statsmodels, what does the Residual Standard Error (RSE) measure?

- A. The percentage of variance explained by the model.
- B. The standard deviation of the response variable Y from the predicted regression line/plane.
- C. The correlation between two predictors.
- D. The total number of parameters in the model.

Correct Answer: B. The standard deviation of the response variable Y from the predicted regression line/plane.

Question 29: What is the difference between a confidence interval and a prediction interval in linear regression?

- A. A confidence interval quantifies uncertainty around the average response $f(X)$, whereas a prediction interval quantifies uncertainty around a single individual response $Y = f(X) + \epsilon$ (incorporating irreducible error).
- B. Prediction intervals are always narrower than confidence intervals.
- C. Confidence intervals account for irreducible error, whereas prediction intervals do not.
- D. Both intervals are identical for large sample sizes n .

Correct Answer: A. A confidence interval quantifies uncertainty around the average response $f(X)$, whereas a prediction interval quantifies uncertainty around a single individual response $Y = f(X) + \epsilon$ (incorporating irreducible error).

Question 30: In Lab 2 (`auto.csv`), when predicting `mpg` using `displacement`, `horsepower`, `weight`, and `acceleration`, which feature exhibited the largest negative coefficient?

- A. `acceleration`
- B. `displacement`
- C. `weight`
- D. `horsepower`

Correct Answer: C. `weight`

Module 3: Regression Diagnostics, Potential Problems & Qualitative Predictors

ISLR 3.3, Video Lectures SL6, SL7, SL9, Lab 2

Question 31: How many dummy variables are required to encode a qualitative predictor variable with $K = 4$ distinct levels (e.g., Region: North, South, East, West)?

- A. 4
- B. 3
- C. 2
- D. 1

Correct Answer: B. 3

Question 32: In dummy variable coding for qualitative predictors, what is the term used for the level that is left out and assigned no dummy variable?

- A. Baseline level
- B. Reference slope
- C. Interaction boundary
- D. High-leverage state

Correct Answer: A. Baseline level

Question 33: Including an interaction term $X_1 X_2$ in a linear regression model relaxes which fundamental assumption?

- A. The linearity assumption
- B. The additivity assumption
- C. The homoscedasticity assumption
- D. The independence of errors assumption

Correct Answer: B. The additivity assumption

Question 34: According to the hierarchical principle in linear regression, when should main effects X_1 and X_2 be retained in a model?

- A. Only if both X_1 and X_2 have p-values less than 0.01.
- B. If an interaction term $X_1 X_2$ is included in the model, even if the individual p-values of X_1 or X_2 are not statistically significant.
- C. Only if the model R^2 is greater than 0.90.
- D. Main effects should always be removed when interaction terms are added.

Correct Answer: B. If an interaction term $X_1 X_2$ is included in the model, even if the individual p-values of X_1 or X_2 are not statistically significant.

Question 35: What diagnostic plot is most effective for detecting non-linearity in a multiple linear regression model?

- A. Plotting target Y against observation index i .
- B. Plotting residuals $e_i = y_i - \hat{y}_i$ versus predicted/fitted values $\hat{y}_i$.
- C. Histogram of predictor X_1 .
- D. Scatterplot of X_1 versus X_2 .

Correct Answer: B. Plotting residuals $e_i = y_i - \hat{y}_i$ versus predicted/fitted values $\hat{y}_i$.

Question 36: What is the major consequence if error terms ϵ_i in a linear regression model are positively correlated (violating independence)?

- A. The OLS coefficient estimates become biased.
- B. The estimated standard errors of the coefficients will underestimate the true standard errors, causing confidence intervals to be narrower than they should be and p-values to be deceptively small.
- C. The training R^2 drops to zero.
- D. The predictors become perfectly collinear.

Correct Answer: B. The estimated standard errors of the coefficients will underestimate the true standard errors, causing confidence intervals to be narrower than they should be and p-values to be deceptively small.

Question 37: How is heteroscedasticity (non-constant error variance) visually identified in a residual plot?

- A. A horizontal straight line at $e_i = 0$.
- B. A funnel or cone shape where the spread of residuals increases or decreases with predicted values $\hat{y}_i$.
- C. A bell-shaped histogram.
- D. A set of discrete vertical parallel lines.

Correct Answer: B. A funnel or cone shape where the spread of residuals increases or decreases with predicted values $\hat{y}_i$.

Question 38: What is a standard mathematical transformation used to mitigate heteroscedasticity in regression data?

- A. Multiplying Y by 100
- B. Applying a log transformation $\log(Y)$ or square root $\sqrt{Y}$ to the response variable
- C. Removing all negative predictor values
- D. Setting the intercept β_0 to zero

Correct Answer: B. Applying a log transformation $\log(Y)$ or square root $\sqrt{Y}$ to the response variable

Question 39: How is an outlier distinguished from a high-leverage point in linear regression?

- A. An outlier has an unusual predictor value X_i , whereas a high-leverage point has an unusual response value Y_i .
- B. An outlier is an observation for which Y_i is far from the value predicted by the model (large studentized residual), whereas a high-leverage point has an unusual predictor value X_i .
- C. Outliers always decrease training R^2 , whereas leverage points increase R^2 .
- D. High-leverage points only occur in simple linear regression.

Correct Answer: B. An outlier is an observation for which Y_i is far from the value predicted by the model (large studentized residual), whereas a high-leverage point has an unusual predictor value X_i .

Question 40: Typically, an observation is flagged as a potential outlier if its studentized residual exceeds what absolute magnitude?

- A. 1
- B. 3
- C. 10
- D. 0.05

Correct Answer: B. 3

Question 41: What is the primary negative impact of severe multicollinearity among predictor variables in a multiple linear regression model?

- A. It causes the model predictions $\hat{Y}$ to become non-linear.
- B. It inflates the standard errors of the estimated coefficients $\hat{\beta}_j$, reducing t-statistics and making it difficult to isolate the individual effect of each predictor.
- C. It reduces training R^2 dramatically.
- D. It prevents the model from computing residuals.

Correct Answer: B. It inflates the standard errors of the estimated coefficients $\hat{\beta}_j$, reducing t-statistics and making it difficult to isolate the individual effect of each predictor.

Question 42: Which metric is used to quantify the severity of multicollinearity for a predictor X_j ?

- A. Residual Sum of Squares (RSS)
- B. Variance Inflation Factor (VIF)
- C. Studentized Residual Ratio
- D. Leverage Statistic (h_i)

Correct Answer: B. Variance Inflation Factor (VIF)

Question 43: As a general rule of thumb, a Variance Inflation Factor (VIF) exceeding what value indicates a problematic level of collinearity?

- A. 1.0
- B. 5 or 10
- C. 0.05
- D. 1000

Correct Answer: B. 5 or 10

Question 44: What are two standard solutions for addressing severe multicollinearity in multiple regression?

- A. Increase model flexibility by adding 10th-degree polynomials.
- B. Drop one of the redundant collinear variables from the model, or combine the collinear variables into a single composite predictor (e.g., via averaging or PCA).
- C. Remove all intercept terms.
- D. Switch from least squares to simple binary thresholding.

Correct Answer: B. Drop one of the redundant collinear variables from the model, or combine the collinear variables into a single composite predictor (e.g., via averaging or PCA).

Question 45: What does a leverage statistic h_i measure for observation i ?

- A. The squared error between y_i and $\hat{y}_i$.
- B. How far the predictor values x_i for observation i deviate from the average predictor values across all observations.
- C. The ratio of R^2 to adjusted R^2 .
- D. The p-value associated with the F-test.

Correct Answer: B. How far the predictor values x_i for observation i deviate from the average predictor values across all observations.

Module 4: Classification Fundamentals & Logistic Regression

ISLR 4.1–4.3, Video Lectures SL12–SL15, Lab 3

Question 46: Why is linear regression ill-suited for qualitative classification tasks with more than two classes (e.g., Stroke=1, Drug Overdose=2, Seizure=3)?

- A. OLS calculations cannot be performed on matrices with more than two columns.
- B. Coding classes as 1, 2, 3 forces an artificial ordering and implies that the distance between Stroke and Overdose equals the distance between Overdose and Seizure.
- C. Classification error cannot be calculated for linear regression.
- D. Linear regression requires all features to be independent.

Correct Answer: B. Coding classes as 1, 2, 3 forces an artificial ordering and implies that the distance between Stroke and Overdose equals the distance between Overdose and Seizure.

Question 47: What is the primary limitation of using linear regression to model a binary outcome (0/1)?

- A. Linear regression cannot handle continuous predictors.
- B. Linear regression can produce predicted probabilities that fall outside the valid $[0, 1]$ interval (i.e., <0 or >1).
- C. Linear regression requires non-parametric tuning parameters.
- D. Linear regression always yields zero training error.

Correct Answer: B. Linear regression can produce predicted probabilities that fall outside the valid $[0, 1]$ interval (i.e., <0 or >1).

Question 48: What is the mathematical form of the logistic (sigmoid) function $p(X)$ used in simple logistic regression?

- A. $p(X) = \beta_0 + \beta_1 X$
- B. $p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$
- C. $p(X) = \log(\beta_0 + \beta_1 X)$
- D. $p(X) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(X - \mu)^2}{2\sigma^2}}$

Correct Answer: B. $p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$

Question 49: How is the term "odds" defined in logistic regression given a probability $p(X) = 0.8$?

- A. $\text{Odds} = 0.8$
- B. $\text{Odds} = \frac{0.8}{1 - 0.8} = 4.0$
- C. $\text{Odds} = 1 - 0.8 = 0.2$
- D. $\text{Odds} = \log(0.8) = -0.22$

Correct Answer: B. $\text{Odds} = \frac{0.8}{1 - 0.8} = 4.0$

Question 50: What transformation of the probability $p(X)$ yields a linear relationship with the predictor variables in logistic regression?

- A. The probability $p(X)$ itself.
- B. The square root $\sqrt{p(X)}$.
- C. The log-odds or logit: $\log\left(\frac{p(X)}{1 - p(X)}\right)$.
- D. The cumulative distribution function of a t-distribution.

Correct Answer: C. The log-odds or logit: $\log\left(\frac{p(X)}{1 - p(X)}\right)$.

Question 51: In a multiple logistic regression model, if the coefficient β_1 for a feature X_1 is 0.693, what is the effect of a one-unit increase in X_1 on the odds of the positive outcome, holding all other features fixed?

- A. The odds increase by 0.693 units.
- B. The odds are multiplied by $e^{0.693} \approx 2.0$ (they double).
- C. The probability increases by 69.3%.
- D. The log-odds decrease by $e^{0.693}$.

Correct Answer: B. The odds are multiplied by $e^{0.693} \approx 2.0$ (they double).

Question 52: What estimation procedure is used to fit the coefficient parameters $\beta_0, \beta_1, \dots, \beta_p$ in logistic regression?

- A. Ordinary Least Squares (OLS)
- B. Maximum Likelihood Estimation (MLE)
- C. Principal Components Analysis (PCA)
- D. Exact algebraic inversion

Correct Answer: B. Maximum Likelihood Estimation (MLE)

Question 53: What statistic in logistic regression output plays the analogous role to the t-statistic in linear regression output?

- A. F-statistic
- B. z-statistic (calculated as $\hat{\beta}_j / \text{SE}(\hat{\beta}_j)$)
- C. Leverage statistic h_i
- D. R^2 score

Correct Answer: B. z-statistic (calculated as $\hat{\beta}_j / \text{SE}(\hat{\beta}_j)$)

Question 54: How does L2 regularization modify the loss function during logistic regression training?

- A. It adds the sum of absolute weight magnitudes $\lambda \sum |\beta_j|$ to the loss.
- B. It appends a penalty proportional to the sum of squared weights $\lambda \sum_{j=1}^p \beta_j^2$ (excluding intercept β_0) to the negative log-likelihood.
- C. It sets all negative coefficients to zero.
- D. It forces the decision threshold to change from 0.5 to 0.1.

Correct Answer: B. It appends a penalty proportional to the sum of squared weights $\lambda \sum_{j=1}^p \beta_j^2$ (excluding intercept β_0) to the negative log-likelihood.

Question 55: In scikit-learn's `LogisticRegression(C=0.1)`, what is the relationship between the parameter `C` and the regularization strength λ ?

- A. `C` is directly proportional to λ ($C = \lambda$).
- B. `C` is the inverse of regularization strength ($C = 1 / \lambda$), so smaller `C` values specify stronger regularization.
- C. `C` represents the number of cross-validation folds.
- D. `C` sets the classification decision threshold.

Correct Answer: B. `C` is the inverse of regularization strength ($C = 1 / \lambda$), so smaller `C` values specify stronger regularization.

Question 56: What happens to model coefficients when strong L2 regularization is applied to a logistic regression model?

- A. Coefficients expand toward positive and negative infinity.
- B. Coefficients shrink toward zero, smoothing the decision boundary and reducing model variance.
- C. Coefficients are set to exactly zero, eliminating features.
- D. Coefficients become uncorrelated with the response.

Correct Answer: B. Coefficients shrink toward zero, smoothing the decision boundary and reducing model variance.

Question 57: How is logistic regression extended to handle multi-class classification ($K > 2$ classes) using the Softmax formulation?

- A. By fitting K separate linear regression models.
- B. By estimating separate coefficient vectors for all K classes and normalizing class probabilities using $P(Y=k|X) = \frac{e^{\beta_{k0}} + \sum \beta_{kj} X_j}{\sum_{l=1}^K e^{\beta_{l0}} + \sum \beta_{lj} X_j}$.
- C. By selecting the median class as a constant prediction.
- D. By setting all non-baseline probabilities to zero.

Correct Answer: B. By estimating separate coefficient vectors for all K classes and normalizing class probabilities using $P(Y=k|X) = \frac{e^{\beta_{k0}} + \sum \beta_{kj} X_j}{\sum_{l=1}^K e^{\beta_{l0}} + \sum \beta_{lj} X_j}$.

Question 58: In a medical screening system where missing a disease case (false negative) is catastrophic, how should the probability decision threshold for logistic regression be adjusted away from the default 0.5?

- A. Lower the threshold (e.g., to 0.2), increasing sensitivity/recall to catch more true positive cases.
- B. Raise the threshold (e.g., to 0.8), reducing false positives.
- C. Set the threshold to 0.0.
- D. Keep the threshold strictly at 0.5 because probability theory forbids changing thresholds.

Correct Answer: A. Lower the threshold (e.g., to 0.2), increasing sensitivity/recall to catch more true positive cases.

Question 59: In Lab 3 ('Wisconsin Breast Cancer'), what effect did applying L2 regularization (' $C=0.1$ ') have on the model coefficients compared to the unregularized baseline?

- A. It increased the magnitude of the coefficients.
- B. It shrank the average absolute magnitude of the coefficients toward zero.
- C. It flipped the signs of all positive coefficients.
- D. It set all 30 feature coefficients to exactly zero.

Correct Answer: B. It shrank the average absolute magnitude of the coefficients toward zero.

Question 60: Under what specific training data condition do unregularized logistic regression maximum likelihood parameter estimates become notoriously unstable?

- A. When n is very large ($n > 100,000$).
- B. When the two classes are perfectly or near-perfectly linearly separable.
- C. When features are standardized to mean 0 and variance 1.
- D. When all features are discrete counts.

Correct Answer: B. When the two classes are perfectly or near-perfectly linearly separable.

Module 5: Generative Models (LDA, QDA, Naive Bayes) & GLMs

ISLR 4.4–4.6, Video Lecture SL16, Lab 3

Question 61: What is the primary conceptual distinction between a discriminative classifier (like Logistic Regression) and a generative classifier (like LDA)?

- A. Discriminative classifiers model $P(X|Y)$, whereas generative classifiers model $P(Y|X)$ directly.
- B. Discriminative classifiers model $P(Y|X)$ directly, whereas generative classifiers model the distribution of predictors within each class $P(X|Y=k)$ and class priors $P(Y=k)$, using Bayes' theorem to derive posterior probabilities.
- C. Generative classifiers can only be used for regression tasks.
- D. Discriminative classifiers require Gaussian feature distributions.

Correct Answer: B. Discriminative classifiers model $P(Y|X)$ directly, whereas generative classifiers model the distribution of predictors within each class $P(X|Y=k)$ and class priors $P(Y=k)$, using Bayes' theorem to derive posterior probabilities.

Question 62: What are the two core mathematical assumptions required by Linear Discriminant Analysis (LDA)?

- A. Features are non-linearly distributed, and classes have unequal variances.
- B. Predictors within each class follow a multivariate Gaussian distribution, and all K classes share a common, identical covariance matrix Σ .
- C. Features are discrete counts following a Poisson distribution, and classes are perfectly separated.
- D. Features are independent of the target, and class prior probabilities are equal to 0.5.

Correct Answer: B. Predictors within each class follow a multivariate Gaussian distribution, and all K classes share a common, identical covariance matrix Σ .

Question 63: Why are the decision boundaries produced by Linear Discriminant Analysis (LDA) strictly linear in x ?

- A. Because LDA uses least squares minimization.
- B. Because the quadratic terms $x^T \Sigma^{-1} x$ cancel out when comparing posterior log-odds between classes due to the shared covariance matrix assumption.
- C. Because LDA sets all off-diagonal covariance terms to zero.
- D. Because LDA forces all class means μ_k to be equal.

Correct Answer: B. Because the quadratic terms $x^T \Sigma^{-1} x$ cancel out when comparing posterior log-odds between classes due to the shared covariance matrix assumption.

Question 64: How does Quadratic Discriminant Analysis (QDA) differ from Linear Discriminant Analysis (LDA)?

- A. QDA assumes a Bernoulli feature distribution, yielding linear boundaries.
- B. QDA assumes each of the K classes has its own distinct covariance matrix Σ_k , resulting in quadratic decision boundaries.
- C. QDA requires feature independence.
- D. QDA uses fewer parameters than LDA.

Correct Answer: B. QDA assumes each of the K classes has its own distinct covariance matrix Σ_k , resulting in quadratic decision boundaries.

Question 65: When is LDA generally preferred over QDA?

- A. When the training sample size n is very small, making variance reduction crucial, and the assumption of a common covariance matrix is reasonable.
- B. When the training sample size n is extremely large and p is small.
- C. When class covariance matrices are known to be wildly different.
- D. When decision boundaries are highly curved.

Correct Answer: A. When the training sample size n is very small, making variance reduction crucial, and the assumption of a common covariance matrix is reasonable.

Question 66: What is the fundamental simplifying assumption made by the Naive Bayes classifier?

- A. The target Y follows a uniform distribution.
- B. Within each class k , the p predictor variables are conditionally independent of one another.
- C. Predictors must be linearly dependent on class labels.
- D. All class means are equal to zero.

Correct Answer: B. Within each class k , the p predictor variables are conditionally independent of one another.

Question 67: What is the major practical benefit of the Naive Bayes feature independence assumption when working with high-dimensional data ($p \gg n$)?

- A. It eliminates the need to estimate joint distribution covariance matrices, drastically reducing parameters and preventing overfitting.
- B. It guarantees 100% classification accuracy.
- C. It allows linear regression to be used for multi-class targets.
- D. It transforms categorical variables into continuous numbers without dummy encoding.

Correct Answer: A. It eliminates the need to estimate joint distribution covariance matrices, drastically reducing parameters and preventing overfitting.

Question 68: When the true feature distributions within each class are strictly multivariate Gaussian with a shared covariance matrix, which method is theoretically optimal?

- A. Logistic Regression
- B. Linear Discriminant Analysis (LDA)
- C. 1-Nearest Neighbor
- D. Decision Trees

Correct Answer: B. Linear Discriminant Analysis (LDA)

Question 69: What type of target response variable is Poisson regression specifically designed to model?

- A. Binary categorical outcomes (0/1).
- B. Unbounded continuous floating-point variables.
- C. Non-negative integer count data (e.g., number of hourly bike rentals, emergency room visits).
- D. Qualitative unordered labels with 5 levels.

Correct Answer: C. Non-negative integer count data (e.g., number of hourly bike rentals, emergency room visits).

Question 70: Why is standard OLS linear regression inadequate for modeling count data (such as the Bikeshare dataset)?

- A. OLS can predict negative counts, violates the constant variance assumption (since variance in count data increases with the mean), and assumes continuous normal errors.
- B. OLS cannot process integer features.
- C. OLS requires multi-class softmax encoding.
- D. Count data cannot be loaded into pandas DataFrames.

Correct Answer: A. OLS can predict negative counts, violates the constant variance assumption (since variance in count data increases with the mean), and assumes continuous normal errors.

Question 71: What is the standard link function used in Poisson regression, and how does it affect coefficient interpretation?

- A. Identity link; a 1-unit increase in X_j adds β_j to Y .
- B. Log link $\log(\lambda) = \beta_0 + \beta_1 X_1$; a 1-unit increase in X_j multiplies the expected mean count $E(Y)$ by a factor of e^{β_j} .
- C. Logit link; a 1-unit increase in X_j doubles the variance.
- D. Softmax link; probabilities sum to zero.

Correct Answer: B. Log link $\log(\lambda) = \beta_0 + \beta_1 X_1$; a 1-unit increase in X_j multiplies the expected mean count $E(Y)$ by a factor of e^{β_j} .

Question 72: In Generalized Linear Models (GLM), what link function is paired with a Bernoulli response distribution to yield Logistic Regression?

- A. Identity link $\eta(\mu) = \mu$
- B. Log link $\eta(\mu) = \log(\mu)$
- C. Logit link $\eta(\mu) = \log\left(\frac{\mu}{1 - \mu}\right)$
- D. Inverse link $\eta(\mu) = 1/\mu$

Correct Answer: C. Logit link $\eta(\mu) = \log\left(\frac{\mu}{1 - \mu}\right)$

Question 73: What phenomenon in count data occurs when the observed variance of the response is substantially higher than the mean, violating pure Poisson model assumptions?

- A. Homoscedasticity
- B. Overdispersion
- C. Multicollinearity
- D. Autocorrelation

Correct Answer: B. Overdispersion

Question 74: How are class prior probabilities π_k typically estimated in LDA when training data is available?

- A. They are always set equally to $1/K$.
- B. By computing the proportion of training observations belonging to each class k : $\hat{\pi}_k = n_k / n$.
- C. By taking the mean of feature matrix X .
- D. Via 10-fold cross-validation grid search.

Correct Answer: B. By computing the proportion of training observations belonging to each class k : $\hat{\pi}_k = n_k / n$.

Question 75: In Lab 3 ('Wisconsin Breast Cancer'), how did Linear Discriminant Analysis (LDA) perform relative to unregularized logistic regression on scaled test data?

- A. LDA failed completely due to non-invertible matrices.
- B. LDA achieved strong classification performance comparable to baseline logistic regression because the underlying feature distributions were well-behaved.
- C. LDA produced quadratic decision boundaries.
- D. LDA required specifying K-neighbors tuning parameters.

Correct Answer: B. LDA achieved strong classification performance comparable to baseline logistic regression because the underlying feature distributions were well-behaved.

Module 6: Resampling Methods, Cross-Validation, Bootstrap & Learning Theory

ISLR 5.1–5.2, Video Lectures SL19, SL30–SL33, Lab 4

Question 76: What are two main limitations of the simple Validation Set approach (e.g., single 50/50 train/test split) for estimating test error?

- A. It cannot be used with logistic regression, and it requires $p > n$.
- B. The test error estimate can be highly variable depending on the random split, and it tends to overestimate the true test error because the model is trained on only half the data.
- C. It requires fitting n models, making it computationally impossible.
- D. It causes severe data leakage.

Correct Answer: B. The test error estimate can be highly variable depending on the random split, and it tends to overestimate the true test error because the model is trained on only half the data.

Question 77: What is the primary computational advantage of k-fold cross-validation (k=5 or 10) over Leave-One-Out Cross-Validation (LOOCV) for complex models?

- A. k-fold CV requires fitting the model only k times rather than n times.
- B. k-fold CV does not require a test dataset.
- C. k-fold CV eliminates all model bias.
- D. k-fold CV uses analytical matrix inversion instead of training.

Correct Answer: A. k-fold CV requires fitting the model only k times rather than n times.

Question 78: Why does Leave-One-Out Cross-Validation (LOOCV) often exhibit higher variance in its test error estimate than k-fold CV (k=10)?

- A. Because LOOCV uses smaller training sets.
- B. Because LOOCV averages the outputs of n models fitted on nearly identical training sets (n-1 overlapping observations), making the outputs highly correlated.
- C. Because LOOCV relies on random shuffling.
- D. Because LOOCV can only be applied to classification problems.

Correct Answer: B. Because LOOCV averages the outputs of n models fitted on nearly identical training sets (n-1 overlapping observations), making the outputs highly correlated.

Question 79: For Ordinary Least Squares linear regression, what "magic formula" allows computing the exact LOOCV error rate by fitting the model only once?

- A. $\text{CV}_{(n)} = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - h_i} \right)^2$, where h_i is the leverage statistic.
- B. $\text{CV}_{(n)} = \text{RSS} \times n$.
- C. $\text{CV}_{(n)} = 1 - R^2$.
- D. $\text{CV}_{(n)} = \frac{\text{TSS}}{p}$.

Correct Answer: A. $\text{CV}_{(n)} = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - h_i} \right)^2$, where h_i is the leverage statistic.

Question 80: How does Stratified k-Fold Cross-Validation improve model evaluation for imbalanced classification datasets?

- A. It multiplies minority class observations by 10.
- B. It ensures that each fold contains approximately the same percentage of target class labels as the complete dataset.
- C. It converts classification into regression.
- D. It randomly drops majority class observations in validation folds.

Correct Answer: B. It ensures that each fold contains approximately the same percentage of target class labels as the complete dataset.

Question 81: What critical rule must be followed during preprocessing (such as feature standardization with `StandardScaler`) to prevent data leakage during cross-validation?

- A. Fit `StandardScaler` on the entire dataset before performing cross-validation splits.
- B. Fit `StandardScaler` strictly on the training folds only within each CV iteration, then transform the validation fold using that fitted scaler.
- C. Standardize only the target variable Y .
- D. Avoid standardizing features when using cross-validation.

Correct Answer: B. Fit `StandardScaler` strictly on the training folds only within each CV iteration, then transform the validation fold using that fitted scaler.

Question 82: What is the core mechanism of the Bootstrap resampling technique?

- A. Partitioning data into k non-overlapping subsets.
- B. Repeatedly drawing random samples of size n with replacement from the original dataset.
- C. Removing outliers using studentized residuals.
- D. Adding Gaussian noise to predictor features.

Correct Answer: B. Repeatedly drawing random samples of size n with replacement from the original dataset.

Question 83: On average, what percentage of original observations are included in a single bootstrap sample of size n (as $n \rightarrow \infty$)?

- A. 50.0%
- B. 63.2% (with approximately 36.8% left out as Out-of-Bag observations)
- C. 100.0%
- D. 95.0%

Correct Answer: B. 63.2% (with approximately 36.8% left out as Out-of-Bag observations)

Question 84: What is a major advantage of using the bootstrap to estimate parameter standard errors compared to classical statistical formulas?

- A. Bootstrap requires zero computation time.
- B. Bootstrap does not rely on rigid parametric assumptions (such as normality of errors or homoscedasticity).
- C. Bootstrap guarantees zero training error.
- D. Bootstrap automatically eliminates multicollinearity.

Correct Answer: B. Bootstrap does not rely on rigid parametric assumptions (such as normality of errors or homoscedasticity).

Question 85: In machine learning model tuning, what is Grid Search?

- A. An algorithm that randomly splits data into continuous intervals.
- B. A systematic approach that evaluates model performance across an exhaustive grid of specified hyperparameter combinations using cross-validation.
- C. A technique for gradient descent optimization.
- D. A method for manual feature selection.

Correct Answer: B. A systematic approach that evaluates model performance across an exhaustive grid of specified hyperparameter combinations using cross-validation.

Question 86: What is the definition of Empirical Risk in statistical learning theory?

- A. The theoretical expected loss over the true, unknown data population distribution.
- B. The average loss computed over a observed, finite training dataset.
- C. The ratio of bias to variance.
- D. The percentage of missing values in a DataFrame.

Correct Answer: B. The average loss computed over a observed, finite training dataset.

Question 87: In statistical learning theory, what does the Vapnik-Chervonenkis (VC) dimension measure?

- A. The number of rows in a dataset.
- B. A model class's flexibility and capacity to fit diverse patterns, defined as the maximum number of points it can shatter.
- C. The computational execution time of gradient descent.
- D. The number of folds in cross-validation.

Correct Answer: B. A model class's flexibility and capacity to fit diverse patterns, defined as the maximum number of points it can shatter.

Question 88: In the Probably Approximately Correct (PAC) learning framework, how does increasing model complexity (higher VC dimension) impact sample complexity?

- A. Higher complexity reduces the required sample size.
- B. Higher complexity increases the quantity of training data required to achieve a desired accuracy guarantee.
- C. Sample complexity is completely independent of model complexity.
- D. Sample complexity drops to zero.

Correct Answer: B. Higher complexity increases the quantity of training data required to achieve a desired accuracy guarantee.

Question 89: In Lab 1 (`california_housing.csv`), what technique was used to identify the optimal K parameter for K-Nearest Neighbors regression?

- A. Evaluating $K=1$ on the training set.
- B. Running 5-fold cross-validation on the training set across candidate K values [1, 3, 5, 7, 9, 15, 25] and picking K with the lowest mean CV RMSE.
- C. Using OLS normal equations.
- D. Plotting ROC curves.

Correct Answer: B. Running 5-fold cross-validation on the training set across candidate K values [1, 3, 5, 7, 9, 15, 25] and picking K with the lowest mean CV RMSE.

Question 90: In Lab 4 (`Diabetes dataset`), what was the purpose of using 5-fold cross-validation when evaluating Ridge and Lasso regularization paths?

- A. To select the optimal regularization parameter λ (or α) that minimizes out-of-sample prediction error.
- B. To compute the training R^2 .
- C. To scale features between 0 and 1.
- D. To convert regression target into binary classification.

Correct Answer: A. To select the optimal regularization parameter λ (or α) that minimizes out-of-sample prediction error.

Module 7: Model Selection & Regularization / Shrinkage

ISLR 6.2, Video Lectures SL15, SL34, Lab 4

Question 91: Why are classical OLS estimates inadequate when the number of features p approaches or exceeds the number of observations n ($p \geq n$)?

- A. OLS cannot compute R^2 .
- B. OLS yields non-unique estimates with extreme variance, perfectly fitting training noise (training $\text{RSS} = 0$) while performing terribly on test data.
- C. OLS forces all slope coefficients to zero.
- D. OLS becomes a non-parametric model.

Correct Answer: B. OLS yields non-unique estimates with extreme variance, perfectly fitting training noise (training $\text{RSS} = 0$) while performing terribly on test data.

Question 92: How does Ridge regression modify the OLS minimization objective?

- A. By adding an L1 penalty $\lambda \sum |\beta_j|$.
- B. By adding an L2 shrinkage penalty $\lambda \sum_{j=1}^p \beta_j^2$ to the Residual Sum of Squares (RSS).
- C. By setting the intercept β_0 to zero.
- D. By dividing RSS by p .

Correct Answer: B. By adding an L2 shrinkage penalty $\lambda \sum_{j=1}^p \beta_j^2$ to the Residual Sum of Squares (RSS).

Question 93: What happens to Ridge regression coefficient estimates $\hat{\beta}_R$ as the tuning parameter λ approaches infinity ($\lambda \rightarrow \infty$)?

- A. Coefficients grow infinitely large.
- B. Coefficients shrink toward zero, approaching the null model (intercept-only), but never become exactly zero.
- C. Coefficients become equal to OLS estimates.
- D. Coefficients fluctuate randomly.

Correct Answer: B. Coefficients shrink toward zero, approaching the null model (intercept-only), but never become exactly zero.

Question 94: Why is feature standardization (e.g., using `StandardScaler`) essential prior to fitting Ridge or Lasso regression models?

- A. Regularization algorithms cannot process non-integer numbers.
- B. OLS requires standardized features.
- C. Unscaled features with large numerical ranges will be artificially penalized more heavily by the L1/L2 penalty terms than features with small ranges.
- D. Standardization guarantees $R^2 = 1.0$.

Correct Answer: C. Unscaled features with large numerical ranges will be artificially penalized more heavily by the L1/L2 penalty terms than features with small ranges.

Question 95: What key property distinguishes the Lasso (L1 regularization) from Ridge regression (L2 regularization)?

- A. Lasso increases model variance.
- B. Lasso uses a square penalty on coefficients.
- C. Lasso's L1 penalty drives coefficient estimates to exactly zero for sufficiently large λ , performing automated feature selection and producing sparse models.
- D. Lasso cannot be tuned using cross-validation.

Correct Answer: C. Lasso's L1 penalty drives coefficient estimates to exactly zero for sufficiently large λ , performing automated feature selection and producing sparse models.

Question 96: Geometrically, why does the Lasso produce sparse solutions (coefficients equal to zero) while Ridge regression does not?

- A. The Lasso constraint region has sharp corners along the coordinate axes where the RSS contours hit first, whereas the Ridge constraint region is a smooth circle/sphere.
- B. Ridge constraint regions are rectangular.
- C. Lasso uses non-Euclidean matrix inversion.
- D. Ridge regression excludes the intercept term.

Correct Answer: A. The Lasso constraint region has sharp corners along the coordinate axes where the RSS contours hit first, whereas the Ridge constraint region is a smooth circle/sphere.

Question 97: In what underlying structural scenario would you expect the Lasso to outperform Ridge regression in test prediction accuracy?

- A. When all p predictors have true non-zero coefficients of similar magnitude.
- B. When a small subset of the p predictors are truly associated with the response, and the remaining predictors are irrelevant noise features.
- C. When $p = 1$.
- D. When features are perfectly collinear.

Correct Answer: B. When a small subset of the p predictors are truly associated with the response, and the remaining predictors are irrelevant noise features.

Question 98: In what structural scenario would Ridge regression generally outperform the Lasso?

- A. When the response is a function of many predictors, all with small-to-moderate non-zero coefficients.
- B. When 99% of predictors have true coefficients equal to zero.
- C. When $n \gg p$.
- D. When no cross-validation is performed.

Correct Answer: A. When the response is a function of many predictors, all with small-to-moderate non-zero coefficients.

Question 99: What is Elastic Net regularization?

- A. A non-parametric tree pruning strategy.
- B. A regularized regression approach that combines both L1 (Lasso) and L2 (Ridge) penalties into a single objective function.
- C. An unregularized Bayesian estimator.
- D. A step function used in splines.

Correct Answer: B. A regularized regression approach that combines both L1 (Lasso) and L2 (Ridge) penalties into a single objective function.

Question 100: Why is Elastic Net often preferred over pure Lasso when working with datasets containing groups of highly correlated features?

- A. Lasso randomly selects only one feature from a group of correlated predictors and discards the rest, whereas Elastic Net's L2 component shrinks correlated features together while L1 maintains sparsity.
- B. Pure Lasso cannot handle correlated features.
- C. Elastic Net reduces computational time to zero.
- D. Pure Lasso requires $p < n$.

Correct Answer: A. Lasso randomly selects only one feature from a group of correlated predictors and discards the rest, whereas Elastic Net's L2 component shrinks correlated features together while L1 maintains sparsity.

Question 101: How is the optimal tuning parameter λ selected in practice for Ridge or Lasso regression?

- A. By picking $\lambda = 0$ always.
- B. By computing cross-validation error across a grid of λ values and selecting the λ that yields the lowest CV MSE (or applying the 1-standard-error rule).
- C. By selecting the λ that maximizes training R^2 .
- D. By setting λ equal to the sample size n .

Correct Answer: B. By computing cross-validation error across a grid of λ values and selecting the λ that yields the lowest CV MSE (or applying the 1-standard-error rule).

Question 102: As the regularization parameter λ increases from 0 to ∞ , how do training RSS, squared bias, and model variance behave?

- A. Training RSS decreases, bias decreases, variance increases.
- B. Training RSS increases, squared bias increases, and model variance decreases.
- C. Training RSS remains constant, bias drops to zero.
- D. All three terms decrease simultaneously.

Correct Answer: B. Training RSS increases, squared bias increases, and model variance decreases.

Question 103: What is the "Curse of Dimensionality" in high-dimensional regression?

- A. Adding more observations increases error.
- B. As the number of features p grows, adding noise features that are unrelated to the response increases model dimensionality and variance, worsening test set error unless regularization is applied.
- C. High-dimensional data cannot be plotted in 2D.
- D. Matrix multiplication fails for $p > 10$.

Correct Answer: B. As the number of features p grows, adding noise features that are unrelated to the response increases model dimensionality and variance, worsening test set error unless regularization is applied.

Question 104: In Lab 4 ('Diabetes dataset'), what happened to the test set MSE when Lasso regularization was applied compared to unregularized OLS regression?

- A. Test MSE increased dramatically due to bias.
- B. Regularization effectively shrank noisy coefficients, yielding lower test MSE and improved generalization over unregularized OLS.
- C. Test MSE remained identical.
- D. The model was unable to fit.

Correct Answer: B. Regularization effectively shrank noisy coefficients, yielding lower test MSE and improved generalization over unregularized OLS.

Question 105: Does Ridge regression perform feature selection?

- A. Yes, it sets half of the coefficients to zero.
- B. No, Ridge regression includes all p predictors in the final model because coefficients are shrunk toward zero but never set to exactly zero.
- C. Yes, if $\lambda > 10$.
- D. Only when features are standardized.

Correct Answer: B. No, Ridge regression includes all p predictors in the final model because coefficients are shrunk toward zero but never set to exactly zero.

Module 8: Non-Linear Models, Splines, GAMs, Decision Trees & Ensembles

ISLR 7, 8, Video Lectures SL10, SL35–SL42

Question 106: Why is polynomial regression $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_d X^d + \epsilon$ still classified as a linear model?

- A. Because its predictions form a straight line.
- B. Because it is linear in its parameters/coefficients β_j , allowing it to be fitted using Ordinary Least Squares on transformed features.
- C. Because it requires binary targets.
- D. Because it has zero bias.

Correct Answer: B. Because it is linear in its parameters/coefficients β_j , allowing it to be fitted using Ordinary Least Squares on transformed features.

Question 107: In regression splines, what are "knots"?

- A. The terminal leaves of a decision tree.
- B. Cutpoints in the range of predictor X where the underlying piecewise polynomial function changes coefficients.
- C. Eigenvalues of the covariance matrix.
- D. Missing observations in a DataFrame.

Correct Answer: B. Cutpoints in the range of predictor X where the underlying piecewise polynomial function changes coefficients.

Question 108: What continuity constraints are enforced on a standard Cubic Spline with K knots?

- A. It must be continuous, and its 1st and 2nd derivatives must also be continuous at each knot.
- B. It must be a straight line everywhere.
- C. Its 3rd derivative must be zero everywhere.
- D. It must have zero variance outside the knots.

Correct Answer: A. It must be continuous, and its 1st and 2nd derivatives must also be continuous at each knot.

Question 109: What is the primary advantage of a Natural Cubic Spline over a standard cubic spline?

- A. Natural cubic splines enforce linear constraints at the boundaries (beyond extreme knots), substantially reducing high variance at the edges of the predictor space.
- B. Natural cubic splines eliminate the need for knots.
- C. Natural cubic splines can only be fit using Lasso.
- D. Natural cubic splines are step functions.

Correct Answer: A. Natural cubic splines enforce linear constraints at the boundaries (beyond extreme knots), substantially reducing high variance at the edges of the predictor space.

Question 110: What is the defining equation of a Generalized Additive Model (GAM) for multiple predictors $X_1, \dots, X_p$?

- A. $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$
- B. $Y = \beta_0 + f_1(X_1) + f_2(X_2) + \dots + f_p(X_p) + \epsilon$, where f_j are smooth non-linear functions.
- C. $Y = \text{sign}(X^T \beta)$
- D. $Y = \min(X_1, X_2)$

Correct Answer: B. $Y = \beta_0 + f_1(X_1) + f_2(X_2) + \dots + f_p(X_p) + \epsilon$, where f_j are smooth non-linear functions.

Question 111: What is the primary split criterion evaluated when growing a Classification Decision Tree?

- A. Residual Sum of Squares (RSS)
- B. Maximizing node purity using Gini Impurity $G = \sum_{k=1}^K p_{mk}(1 - p_{mk})$ or Cross-Entropy
- C. Minimizing the number of features
- D. Maximizing leverage h_i

Correct Answer: B. Maximizing node purity using Gini Impurity $G = \sum_{k=1}^K p_{mk}(1 - p_{mk})$ or Cross-Entropy

Question 112: Which of the following is a major advantage of decision trees over linear regression models?

- A. Decision trees easily fit diagonal, smooth linear boundaries with low variance.
- B. Decision trees natively handle non-linear relationships, qualitative features without dummy coding, and are invariant to monotonic feature transformations (no scaling needed).
- C. Decision trees are immune to overfitting.
- D. Decision trees always have higher prediction accuracy than random forests.

Correct Answer: B. Decision trees natively handle non-linear relationships, qualitative features without dummy coding, and are invariant to monotonic feature transformations (no scaling needed).

Question 113: How does Cost-Complexity Pruning (Weakest Link Pruning) prevent decision trees from overfitting?

- A. By setting the maximum tree depth to 1.
- B. By growing a large initial tree T_0 and collapsing subtrees that yield minimal decrease in error relative to a complexity penalty $\alpha |T|$, using CV to select α .
- C. By replacing classification trees with linear regression.
- D. By eliminating feature scaling.

Correct Answer: B. By growing a large initial tree T_0 and collapsing subtrees that yield minimal decrease in error relative to a complexity penalty $\alpha |T|$, using CV to select α .

Question 114: What is the primary statistical mechanism by which Bagging (Bootstrap Aggregation) improves model performance over a single decision tree?

- A. It reduces model bias by fitting 100th-degree polynomials.
- B. It reduces model variance by generating B bootstrapped training sets, fitting deep unpruned trees to each, and averaging their predictions.
- C. It performs feature selection by setting weights to zero.
- D. It forces all split points to be identical.

Correct Answer: B. It reduces model variance by generating B bootstrapped training sets, fitting deep unpruned trees to each, and averaging their predictions.

Question 115: How do Random Forests modify standard Bagging to decorrelate individual decision trees?

- A. By forcing all trees to use the same root feature.
- B. At each split, the algorithm is restricted to considering a random subset of $m \approx \sqrt{p}$ features rather than all p features.
- C. By training trees sequentially on residuals.
- D. By using K-means clustering on the features.

Correct Answer: B. At each split, the algorithm is restricted to considering a random subset of $m \approx \sqrt{p}$ features rather than all p features.

Question 116: What is the primary operational difference between Random Forests and Gradient Boosting?

- A. Random Forests build deep trees independently in parallel to reduce variance, whereas Gradient Boosting builds shallow trees sequentially to fit residuals and reduce bias.
- B. Random Forests fit residuals, whereas Gradient Boosting averages independent trees.
- C. Random Forests cannot be used for classification.
- D. Gradient Boosting does not use decision trees.

Correct Answer: A. Random Forests build deep trees independently in parallel to reduce variance, whereas Gradient Boosting builds shallow trees sequentially to fit residuals and reduce bias.

Question 117: In Gradient Boosting, what role does the learning rate / shrinkage parameter λ (or `learning_rate`) play?

- A. It sets the maximum number of features per split.
- B. It controls the rate at which boosting learns by scaling the contribution of each new tree ($0 < \lambda \leq 1$), where smaller values require more trees but prevent overfitting.
- C. It sets the test set split ratio.
- D. It specifies the Gini threshold.

Correct Answer: B. It controls the rate at which boosting learns by scaling the contribution of each new tree ($0 < \lambda \leq 1$), where smaller values require more trees but prevent overfitting.

Question 118: How does Out-of-Bag (OOB) error estimation work in Bagging and Random Forests?

- A. By testing the ensemble on synthetic data.
- B. For each observation i , an OOB prediction is made using only the trees for which observation i was left out of the bootstrap sample (approx 36.8% of trees), yielding an unbiased validation score without cross-validation.
- C. By computing training MSE on K-fold splits.
- D. By dropping 50% of predictors.

Correct Answer: B. For each observation i , an OOB prediction is made using only the trees for which observation i was left out of the bootstrap sample (approx 36.8% of trees), yielding an unbiased validation score without cross-validation.

Question 119: What information is displayed in a Partial Dependence Plot (PDP) for a feature X_j ?

- A. The local variation of a single individual observation i .
- B. The global marginal effect of feature X_j on the model's predicted outcome, averaged across all instances and other features in the dataset.
- C. The correlation matrix of predictors.
- D. The ROC AUC curve.

Correct Answer: B. The global marginal effect of feature X_j on the model's predicted outcome, averaged across all instances and other features in the dataset.

Question 120: How do Individual Conditional Expectation (ICE) plots differ from Partial Dependence Plots (PDPs)?

- A. ICE plots display R^2 scores.
- B. PDPs show the global average feature effect, whereas ICE plots show individual curves for each separate instance, revealing local variations and individual interaction effects.
- C. ICE plots can only be drawn for simple linear regression.
- D. PDPs require feature standardization, whereas ICE plots do not.

Correct Answer: B. PDPs show the global average feature effect, whereas ICE plots show individual curves for each separate instance, revealing local variations and individual interaction effects.